

Harassment Isn't Virtual When It Feels Real: Understanding Emotional Impact of Gendered Embodiment in VR

Category: Research

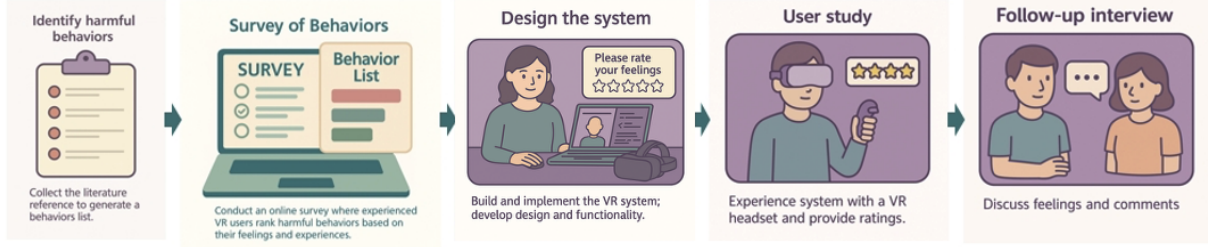


Figure 1: **Workflow Overview.** Five-step study workflow: from identifying harmful behaviors to follow-up interviews. The cartoons are generated by AI.

ABSTRACT

Harassment in Social Virtual Reality (SVR) can feel more intense than in traditional online platforms due to embodiment and real-time interaction. However, how gender cues shape users' perception of harassment in immersive environments remains unclear. We investigate how avatar gender, harasser gender, and participant gender influence perceived discomfort during social VR interactions. Using a customized Unity-based VR environment, we present participants with harassment scenarios enacted by non-player characters representing female, male, and neutral genders. The behaviors are derived from a prior online survey with 61 respondents. In a mixed-methods user study ($N=46$), participants experienced seven predefined harassment behaviors and reported their emotional discomfort on a 5-point Likert scale after each interaction, followed by post-study interviews. Results show that gender cues influence how harassment is perceived in immersive environments, highlighting the role of embodiment in shaping users' emotional responses. We contribute empirical evidence on gendered harassment in SVR and discuss implications for mitigation strategies, avatar design, and ethical safeguards in immersive social systems.

Index Terms: Social Virtual Reality, Gendered Harassment, User Experience

1 INTRODUCTION

Harassment remains a deeply pervasive social problem around the world. In the United States, someone is sexually assaulted every 68 seconds [42]. Nearly 44% of adult women and 25% of adult men in the U.S. have experienced unwanted sexual contact in their lifetime, and one in three women and one in 33 men in the U.S. have experienced an attempted or completed rape in their lifetime [20]. A 2024 United States nationwide study reported that 82% of women and 42% of men reported experiencing some form of sexual harassment or assault in their lifetime [60]. These numbers demonstrate the scale of gendered violence across populations. Online platforms have created new spaces for harassment. A 2021 Pew survey found that 41% of U.S. adults experienced some form of online harassment, and about 25% faced more severe types such as threats, stalking, sustained abuse, or sexual harassment [64]. Data from the Women's Media Center [69] show similar patterns. Between 25% and 40% of Internet users report harassment, with 45% encountering more severe forms. Women comprise 57% of harassment victims, and users with female usernames receive sexually explicit or threatening messages 25 times more often than others. These findings show that gender plays a strong role in online harassment.

Immersive virtual reality (VR) environments intensify these risks. VR systems support embodiment and real-time interaction. As a result, harassment in VR can feel more direct and invasive than in traditional online platform. A survey of social VR users found that 49% of women experienced sexual harassment, including unwanted touching and stalking, while men reported verbal abuse and threats [57]. Qualitative research further highlights that embodied harassment in VR can feel physically threatening, especially when avatars resemble users' real-life identities [51].

Despite these insights, critical gaps remain. Prior studies have mainly reported prevalence rates or described individual experiences [12, 9, 72, 47]. However, fewer studies have examined how identity-related design factors shape user discomfort. In particular, little is known about how participant gender, avatar gender, and harasser gender influence emotional responses in immersive environments. It is also unclear how these identity factors interact. These interactions may increase or reduce perceived discomfort. Understanding these dynamics is essential for designing safer social VR platforms. To address these gaps, in this study, we investigate two research questions: **RQ1:** How do participant gender, avatar gender, and harasser gender influence user discomfort in immersive social VR? **RQ2:** How do the combinations among participant gender, avatar gender, and harasser gender shape users' emotional responses in SVR?

To explore these questions, we conducted a within-subjects mixed-methods study in a custom-built VR environment. Participants ($N = 46$) experienced seven predefined harassment behaviors, which were determined by a multi-participant online survey ($N = 61$). As shown in Fig. 1, our research followed a five-step process: First, we reviewed prior literature to identify harmful behaviors in VR. Second, we surveyed experienced VR users to evaluate these behaviors. Third, we implemented a controlled VR environment with male, female, and neutral avatars. Fourth, we conducted a user study in which participants experienced these behaviors and rated their discomfort in real time using VR headsets. Fifth, we conducted follow-up interviews to collect qualitative feedback about emotional reactions and perceptions. Our analysis reveals three key findings. First, participant embodiment influenced discomfort ratings. Participants who embodied female avatars reported higher overall discomfort. Second, harasser gender presentation significantly affected emotional responses. Male-presenting harassers elicited stronger negative reactions across participants. Third, interaction effects emerged across identity factors, indicating that gender dynamics operate jointly rather than independently in shaping user experience. These results

show that identity cues in immersive environments play a critical role in how harassment is experienced. These findings suggest that social VR platforms should consider gender dynamics when designing safety mechanisms and moderation systems.

The remainder of the paper is organized as follows. We present background on harassment and social VR, and follow-up works in Section 2. We describe study design and implementation details in Section 3. Section 4 presents the quantitative and qualitative results, and Section 5 and Section 6 discuss the insights based on the hypothesis and limitations. We make concluding remarks in Section 8.

2 RELATED WORK

2.1 Harassment in Social virtual reality

In the domain of online and virtual harassment, researchers have explored various facets of user safety, platform moderation, and the role of emerging technologies, particularly in immersive environments like Social Virtual Reality (SVR). Harassment in these spaces has been studied in-depth to identify the unique challenges posed by their interactive and immersive nature.

Harassment in Social VR environments is a recurring concern, with studies showing that the embodied and immersive qualities of these platforms magnify users' experiences of abuse. For example, users report that harassment in VR feels more personal and invasive due to the physicality of avatars and the sense of presence VR provides [13, 38]. The study reveals that harassment in VR differs significantly from traditional online harassment due to the sense of physical presence that users experience through their avatars. Embodied harassment, where users experience violations of their virtual bodies, often leaves a more substantial emotional impact than other forms of abuse. Since VR is still in its early stages of development, shared norms of appropriate behavior have not yet been fully established, which makes it challenging to regulate incidents using conventional moderation systems. Research indicates that current tools and governance strategies frequently fail to address the distinct challenges of harassment in VR, as traditional reporting mechanisms often overlook the intensity and embodied nature of these experiences.

Research has also expanded to examine how women, in particular, navigate harassment in VR environments, employing strategies such as blocking or avoiding specific spaces. Schulenberg et al. [52] measured the strategies women use to cope with harassment, such as blocking users or avoiding specific VR spaces altogether. The study highlights the limitations of existing moderation tools, calling for platform improvements that provide gender-sensitive protections and mechanisms for reporting harassment. The researchers argue that gender-specific design interventions are necessary to create safer user environments. Findings suggest that platform designs must evolve to account for gendered experiences of harassment, offering more robust protections and adaptive moderation systems. Furthermore, AI-based moderation is viewed as a promising solution for managing harassment in VR. The challenges of AI in moderation, particularly its inability to understand context or nuance, are echoed across multiple studies, with calls for a hybrid approach that blends automated tools with user-driven reporting. Another idea of using AI to mitigate harassment is proposed by Schulenberg et al. [53]. This paper discusses how AI can be integrated into VR platforms to detect harmful behavior in real-time, but also acknowledges the limitations of AI in fully understanding the context of social interactions. The authors propose a hybrid approach that combines AI-based systems with human moderators to assess complex social cues and nuanced behaviors accurately. This method aims to provide more efficient and scalable moderation in the social VR space.

Harassment in online environments disproportionately affects marginalized groups and content creators, who face distinct chal-

lenges on platforms like YouTube and TikTok. The paper [59] delves into the persistent abuse these creators endure, ranging from toxic comments to more severe issues like stalking and doxing. The study emphasizes the need for platforms to develop more robust safety measures and user support systems to effectively mitigate harassment and protect users from harm. In a related study, researchers investigate how harassment behaviors vary across different platforms, revealing the importance of context in understanding what constitutes harassment [55]. Behavior considered acceptable in one online community could be deemed offensive in another. This work suggests that a one-size-fits-all approach to harassment moderation is ineffective and calls for governance strategies that consider the specific social norms and expectations of each community [56]. SoK [58] provides a comprehensive overview of how online harassment has evolved, examining the broader landscape of hate speech and harassment across platforms. This paper examines the tension between free expression and user safety, highlighting the need for adaptive technical, design, and policy interventions to address harassment in the evolving landscape of online abuse. These studies collectively highlight the complexity of addressing harassment in both virtual and online environments. As online abuse continues to evolve, platform governance, moderation tools, and safety measures must keep pace to ensure users are effectively safeguarded from harassment [13, 38, 52, 53, 58].

2.2 Online harassment and Identity Dynamics

Before 1984, researchers began studying online harassment [64, 27]. They describe how anonymity and reduced social cues in digital environments facilitate flaming, abuse, and early forms of online harassment. [31] Till today, online harassment still is one of the most crucial issues affecting both professional and social environments, as digital platforms enable rapid and often anonymous escalation of harmful behavior. While harassment has been frequently discussed in the context of employment, its influence extends to social media, forums, virtual workspaces, and everyday online interactions. In the digital age, harassment is more pervasive and complex, making it an increasing focus for legal experts, psychologists, and social scientists. Researchers and professionals categorize online harassment into various forms, such as verbal, visual, and psychological [29]. For example, after investigating 25 users, a study [12] described three possible harassment behaviors, including verbal harassment that personal insults or hateful words, which require communication between users, including text messages and oral harassment, physical harassment would simulate touch or grabbing, including sexual gestures and environment harassment always display graphic content on a shared screen.

These categories provide a broader understanding of harassment and facilitate the development of more targeted interventions for prevention and response. Experts in law and psychology have refined the definition of online harassment to reflect the distinct challenges presented by digital platforms. While harassment in physical spaces is more recognizable, the online environment allows harassment to take on new forms, including cyberbullying, coordinated attacks, and stalking, which can be as harmful as physical harassment [25]. Online platforms, such as social media, gaming environments, and forums, are frequent arenas for this behavior, where individuals can be targeted based on various aspects of their identity [21]. According to the statistical results, Online harassment is no less harmful because it occurs online. Virtual harassment can leave lasting harm, as the content is often public, unerasable, and just as emotionally damaging as in-person harassment [30]. Forty-one percent of Americans have been personally subjected to harassing behavior online, and a larger share, around 66%, have witnessed these behaviors directed at others [64]. People from historically oppressed groups are more likely to be harassed online, and that harassment is likely to be more severe [27]. Online harassment mir-

rors the inequalities we see in society, often including racist, sexist, transphobic, homophobic, ableist, or other hate speech based on aspects of someone's identity. Victims of online harassment often have little recourse. Due to the anonymity of the internet, many victims have no idea who is behind the harassment and have few ways to protect him/herself, even if they do know the person causing harm [15]. Though legal frameworks are essential for protecting individuals from online harm, there is often a gap between offensive behavior and behavior that constitutes illegal harassment [61].

Not all online offenses meet the legal threshold for harassment: U.S. statutes and case law generally require either a course of conduct or "severe or pervasive" behavior, meaning that minor slights, insults, or isolated incidents are not illegal harassment unless part of a repeated or extreme pattern [2, 6, 7, 8, 3, 4, 5, 1, 16, 44, 63]. However, when online harassment significantly disrupts a person's personal or professional life, it crosses the threshold into unlawful behavior. Effectively addressing online harassment requires building safe and inclusive digital environments for both direct and indirect victims through continuously evolving platform policies, user protections, and legal responses that keep pace with the dynamic nature of digital communication [17, 48, 28, 54, 18]. As society becomes increasingly reliant on digital spaces, harassment policies must continue to evolve, focusing on prevention and timely intervention to tackle the growing challenges of online harassment.

In recent years, numerous research approaches have focused on harassment in online environments. Wei et al. [68] interviewed subject matter experts to assess the effectiveness of existing safety tools and emphasized the need for stronger, platform-level interventions. Other studies have approached the issue from different perspectives, including the design of reporting systems, the role of moderation practices, the psychological impact of abuse, and the influence of identity on user experiences. Collectively, this body of work highlights the complex nature of online harassment and the ongoing need for adaptive protections [22, 53, 36, 52, 37].

3 STUDY DESIGN AND IMPLEMENTATION

This section outlines the design and implementation of our study. As illustrated in Fig. 2, the research process consisted of four main stages. We first identified and validated a set of harassment behaviors through a literature review and an online survey of experienced VR users. Next, we implemented a custom social VR environment that included these behaviors while ensuring both control and realism. Then, we recruited participants and conducted a controlled user study, combining quantitative discomfort ratings with qualitative interviews. Finally, we analyzed the quantitative results and qualitative data to derive key insights. The following subsections describe these stages in detail.

3.1 Survey of Harassment Behaviors in Social VR environments

Harassment is a well-documented issue across legal frameworks, academic research, and digital communities. Law enforcement agencies and legal service providers publish public resources that identify various forms of harassment and support victims of abuse [29, 14]. These materials describe harmful behaviors ranging from physical threats to sexual misconduct. Academic institutions and researchers have also examined harassment in digital contexts. For example, Harvard Law School provides extensive documentation and guidance on harassment cases [50], and prior work in human-computer interaction and digital ethics has investigated harassment in online and virtual environments [22, 52].

Based on these sources, we compiled a list of commonly cited physical harassment behaviors documented in legal, academic, and advocacy literature. These behaviors include *middle-finger gestures, kicking, taunting, hitting, shaking fists angrily, making loser gestures, unwelcome touching of hair or clothing, inappropriate*

hugging, direct threats of harm, leering or suggestive facial expressions, touching someone's breast, and deliberately brushing or rubbing one's body against another. These actions are frequently reported as indicators of serious harassment [29, 14, 49, 26, 43].

To validate and contextualize this list in social VR, we conducted an online survey using Qualtrics [46] with 61 VR headset users to understand public perceptions of the most harmful forms of harassment in social VR environments. Participants included 30 females and 31 males, with over 50 reporting prior VR experience and 42 indicating prior use of social VR platforms.

Fig. 3 presents the rate of participants who evaluated the severity of various harassment behaviors in VR. The top seven behaviors are selected by over 40% of respondents, representing the most widely recognized forms of harassment in Social VR. These behaviors include *Touching someone's breast, Threats of harm, Rubbing body, Unwanted touching, Hitting, Taunting, and Kicking*. Tab. 1 presents the harassment behaviors we adopted.

3.2 Social VR environment design

Our study was conducted in a custom-built social VR environment developed using Ubiq [23]. This open-source platform provides foundational features for multi-user VR systems, including connection management, basic movement, and avatar support. The environment was deployed on Meta Quest 2 headsets [39] to facilitate immersive interaction during the user study. Ubiq was chosen for its functional interactions as a real social VR platform and its integration with Unity [62], enabling the flexibility for add-on features and control over interactive behaviors.

The virtual environment was modeled as a stylized art museum, featuring paintings and sculptures to create a neutral, immersive setting for participant movement and interaction. Harassment behaviors were simulated using animated non-player characters (NPCs), with gestures and motion clips sourced from Mixamo [10]. To highlight the gender-sensitive purpose, we include three common avatar gender categories, female, male, and neutral, for both harasser avatars and participants' avatars. Avatars were designed in a cartoon-like style and programmed accordingly. Each harasser avatar was assigned one of seven predefined harassment behaviors, ensuring that no avatar repeated a behavior. This design allowed us to isolate the effects of each behavior without cross-contamination between avatars.

Participant responses were collected through an integrated rating interface that appeared after they exited a predefined proximity area around each harasser. These proximity zones were calibrated through pilot testing to reflect interpersonal distances likely to influence user perception in immersive settings. After each interaction, participants rated their emotional discomfort using a 5-point Likert-type scale [34]. All ratings were saved locally in JSON format for later analysis.

To minimize spatial bias, we created three identical versions of the museum layout, each with randomized avatar positions. To enhance realism and ecological validity, we included two passive tourist NPCs in the museum to simulate the presence of other visitors viewing the exhibits. These background agents contributed to a more lifelike social VR environment without interfering with the core harassment interactions. A limitation of this design is that harassment was simulated by NPCs rather than enacted through live human interaction. We made this choice for two reasons. First, to ensure that each harassment behavior was presented in precisely the same way across participants, avoiding individual variation or bias. Second, to reduce potential emotional impact, participants were informed after the study that the harassers were NPCs. Fig. 4 illustrates the simulated social VR environment. Subfigure (a) presents a top-down view of the entire landscape, where white blocks near the avatars indicate sensing areas that trigger specific behaviors. Subfigure (b) depicts the entrance of the environment, while sub-

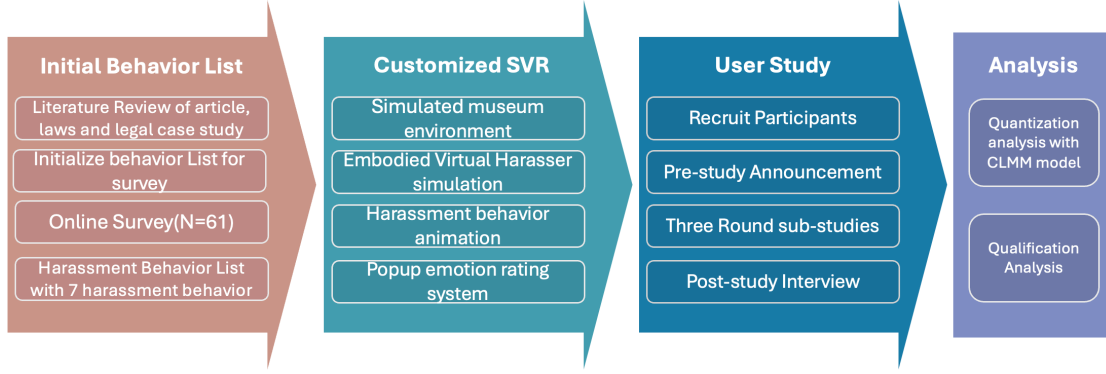


Figure 2: **Overview of the study pipeline.** Seven harassment behaviors were grounded through prior literature and an online survey, implemented in a customized SVR environment with animated agents, and evaluated through a controlled multi-round user study. Quantitative and qualitative analyses were conducted to assess user discomfort and experiential responses.

Table 1: **Harassment Behavior List** sexual & non-sexual w/ reference

Harassment Behavior	Harassment type	source
Taunting others	Non-sexual Harassment	[14, 49]
Kicking others	Non-sexual Harassment	[14, 26]
Hitting others	Non-sexual Harassment	[14, 26]
Direct threats of intent to inflict harm	Non-sexual Harassment	[14, 49, 43]
Touching someone's breast intentionally	Sexual Harassment	[29, 43]
Kissing or hugging someone inappropriately	Sexual Harassment	[29, 49]
Deliberately brushing or rubbing your body against someone	Sexual Harassment	[29, 43]

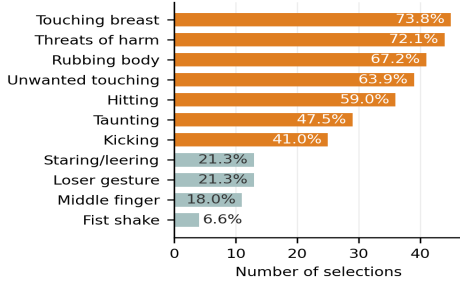


Figure 3: **Online Survey result of harassment behaviors List in social VR.** Each bar represents the number and percentage of participants ($N = 61$) who selected the seven most harmful or severe behaviors with 'orange' bars. Only behaviors selected by more than 40% of participants are considered among the most concerning forms of harassment in immersive virtual environments.

figure (c) highlights the integrated rating system used to capture participants' feedback after interactions.

3.3 Participants and study design

We recruited 46 participants for this study. All participants self-identified within the gender binary, with 23 identifying as male and 23 as female. Most participants were university students ($N = 36$), while others were employed in industry or self-employed ($N = 8$), or currently unemployed ($N = 2$). Participants came from diverse racial backgrounds, including 18 Asian, 17 White, 6 Latino, and 5 Black or African American individuals. Ages ranged from 18 to 45, with the majority under 20, followed by 12 between 20 and 30, and 5 over 30. Participants were recruited through on-campus posters and flyers and received a \$20 Amazon gift card as compensation.

We employed a within-subjects design in which each participant experienced 21 harassment scenarios: seven unique behav-

iors, each enacted by avatars of three different gender types (female, male, neutral). These events were distributed across three museum scenes, with a unique spatial layout of NPCs. To mitigate order and location effects, participants would randomly choose one out of three versions for each round study while ensuring exposure to the same set of harassment scenarios.

Each study session lasted approximately one hour and consisted of three VR rounds followed by a 15- to 20-minute semi-structured interview. In each round, participants were assigned a different avatar gender from a predefined list (female, male, or neutral) and instructed to traverse a virtual gallery. As participants approached each avatar, a harassment animation was automatically triggered. Upon exiting the interaction zone, a rating interface appeared, prompting participants to report their discomfort on a 5-point Likert-type scale (0 = no discomfort, 4 = extremely uncomfortable/would like to leave). Each round lasted approximately 10–15 minutes. After completing all three rounds, participants took part in a post-study interview. The interview explored the most distressing behaviors and emotional changes experienced during the study, as well as open-ended questions about how avatar gender influenced these experiences, along with any suggestions or concerns they wished to share.

This study was reviewed and approved by the Institutional Review Board (IRB). All participants provided informed consent before participation.

3.4 Statistical Analysis

To evaluate the proposed hypotheses, we adopted an estimand-driven analysis strategy based on the Cumulative Link Mixed-effects Model (CLMM) [11]. This approach is appropriate for ordinal outcome data and ensures conceptual alignment between model specification, hypothesis formulation, and reported effects. All hypotheses were evaluated using a single full factorial model and are reported on the response scale, with 95% confidence intervals, which can avoid inconsistencies that can arise from fitting separate additive and interaction models.

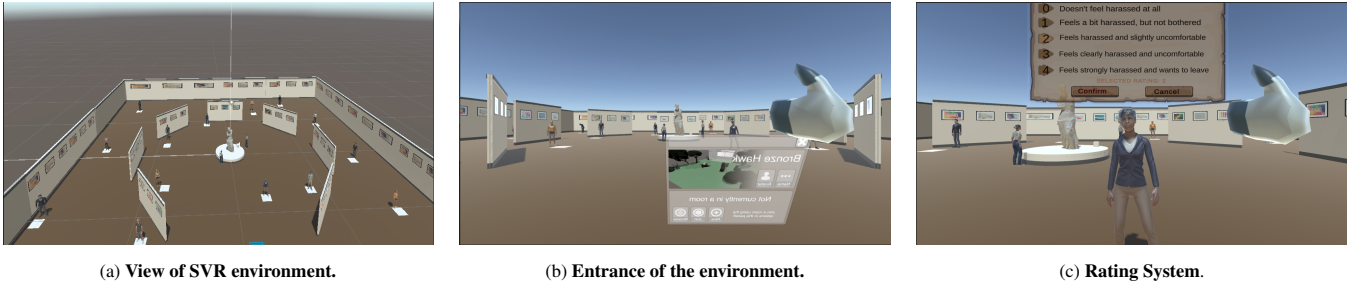


Figure 4: Simulated Social VR (SVR) environment with Rating System.

The dependent variable was participants' reported discomfort, measured on a 5-point Likert scale ranging from 0 (no discomfort) to 4 (extreme discomfort). Fixed effects included participant gender (male, female), avatar gender (female, male, neutral), and virtual harasser gender (female, male, neutral). Participant identity was modeled as a random intercept to account for repeated measurements across experimental conditions. The order of virtual harasser appearance (left-to-right or right-to-left) was randomized and treated as a control variable. An overview of all experimental variables is provided in Tab. 2.

Because different hypotheses target different estimands, we employed distinct but complementary evaluation strategies within the same CLMM. Hypotheses concerning main (marginal) effects were evaluated using contrasts of predicted cumulative probabilities of discomfort, derived from the full model and averaged across the remaining experimental factors. These contrasts estimate population-level differences in the likelihood of reporting a given or higher level of discomfort, independent of specific reference categories. Hypotheses concerning interaction effects were evaluated directly using tests of the corresponding interaction terms in the CLMM. All statistical significance tests were conducted using Wald z -tests. For all analyses, we report model-derived effect estimates, standard errors, and p -values, and visualize predicted cumulative probabilities with 95% confidence intervals. Effect sizes are reported as odds ratios (OR), where $OR > 1$ indicates increased likelihood of reporting higher discomfort and $OR < 1$ indicates decreased likelihood.

3.5 Research Question and Hypothesis

This study examines how gender-related factors influence perceived discomfort with harassment in Social Virtual Reality (SVR). We focus on three variables: participant gender (P), embodied avatar gender (A), and virtual harasser gender (H). Based on these factors, we formulate the following research questions and hypotheses. **RQ1:** How do participant gender, avatar gender, and harasser gender influence user discomfort in immersive social VR? To address this question, we examine the following effects: **H1** participant gender would influence perceived discomfort, with female participants exhibiting higher predicted discomfort than male participants across conditions. **H2** avatar gender would affect perceived discomfort, such that embodying a female avatar would be associated with higher predicted discomfort than embodying male or neutral avatars. **H3** virtual harasser gender would influence perceived discomfort, with gendered virtual harassers eliciting higher predicted discomfort than neutral virtual harassers across participant and avatar conditions.

RQ2: How do the combinations among participant gender, avatar gender, and harasser gender shape users' emotional responses in SVR? For this research question, we considered the two-way interactions, including **H4** harasser's gender and participant's gender, **H5** avatar gender and participant's gender, **H6** avatar gender with harasser gender, and **H7** three-way interactions.

4 EVALUATION AND RESULTS

We first report the main effects of participant gender, avatar gender, and harasser gender on perceived discomfort (Sec. 4.1). We then present the interaction effects examining how combinations of these gender factors influence discomfort in social VR (Sec. 4.2).

4.1 Discomfort Across Participant, Avatar, and Harasser Gender

We examined the main effects of participant gender, avatar gender, and virtual harasser gender on reported discomfort. These hypotheses concern overall differences in predicted discomfort averaged across the remaining experimental factors. All main effects were evaluated using contrasts of predicted cumulative probabilities of reporting higher discomfort, derived from the cumulative link mixed-effects model and averaged across the remaining factors. Fig. 5 summarizes these effects and Tab. 3 indicates the detailed statistical results. For brevity, we use the abbreviations P, A, and H to denote participant gender, avatar gender, and virtual harasser gender.

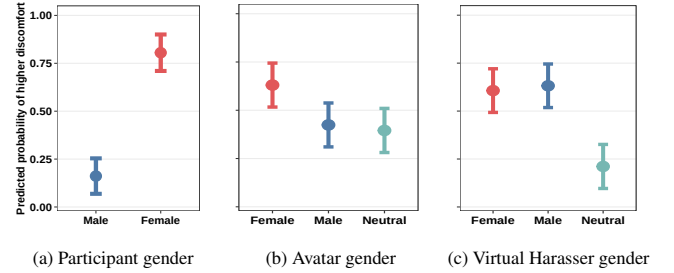


Figure 5: Main effects on predicted probability of higher discomfort. Model-derived predicted cumulative probabilities of reporting higher discomfort by (a) participant gender, (b) avatar gender, and (c) virtual harasser gender. Estimates are averaged over remaining factors; error bars indicate 95% confidence intervals.

4.1.1 Participant Gender Influences Perceived Discomfort (H1)

We hypothesized that participant gender would influence perceived discomfort, with female participants exhibiting higher predicted discomfort than male participants across conditions.

As shown in Fig. 5a, participant gender exhibited a strong and statistically significant main effect on predicted discomfort. Averaged across avatar gender and virtual harasser gender, female participants showed a substantially higher predicted cumulative probability of reporting high discomfort than male participants (contrast: Male–Female = -0.633 , $SE = 0.067$, $z = -9.60$, $p < 0.0001$). The estimated marginal predicted cumulative probability was 0.805 (95% CI [0.710, 0.900]) for female participants, and 0.161 (95% CI

Table 2: **Experimental Variable Overview.** IV: Independent Variable. DV: Dependent Variable.

Factor	Variable Type	Description
Participant Gender	Between-subjects IV	Male, Female
Avatar Gender	Within-subjects IV	Female, Male, Neutral
Virtual Harasser Gender	Within-subjects IV	Female, Male, Neutral
Order of Harasser Appearance	Control (Randomized)	Left-to-right or Right-to-left
Discomfort Rating	Dependent Variable (DV)	0–4 scale: Participant’s reported level of discomfort in response to harassment (0 = not uncomfortable at all, 4 = extremely uncomfortable)

Table 3: **Main effects on predicted discomfort.** Pairwise contrasts of marginal predicted cumulative probabilities derived from the cumulative link mixed-effects model. Effect sizes are reported as odds ratios (OR). Results are averaged across the remaining experimental factors.

Hyp.	Comparison	OR	z	p
H1 (P)	Male – Female	0.53	–9.60	< 0.001
H2 (A)	Female – Male	1.23	2.54	< 0.05
	Female – Neutral	1.26	2.89	< 0.05
	Male – Neutral	1.03	0.35	> 0.05
H3 (H)	Female – Male	0.98	–0.31	> 0.05
	Female – Neutral	1.49	4.82	< 0.001
	Male – Neutral	1.52	5.13	< 0.001

[0.069, 0.254]) for male participants. The effect size was large (OR = 0.53, where OR = 1.90 when expressed as Female vs. Male), indicating that female participants were substantially more likely to report higher discomfort. These results support the hypothesis that female participants experience higher discomfort than male participants across conditions.

4.1.2 Embodied Avatar Gender Modulates Discomfort (H2)

We hypothesized that avatar gender would affect perceived discomfort, such that embodying a female avatar would be associated with higher predicted discomfort than embodying male or neutral avatars.

Avatar gender exhibited a statistically significant main effect on predicted discomfort, shown in Fig. 5b. When averaged across participant gender and virtual harasser gender, the female avatar condition yielded a higher predicted cumulative probability of high discomfort than both the male avatar condition (contrast: Female–Male = 0.206, SE = 0.081, $z = 2.542$, $p < 0.05$) and the neutral avatar condition (contrast: Female–Neutral = 0.235, SE = 0.082, $z = 2.89$, $p < 0.05$, OR = 1.23). No statistically reliable difference was observed between the male and neutral avatar conditions ($p > 0.05$). Corresponding predicted cumulative probabilities were 0.630 (95% CI [0.517, 0.744], OR = 1.26) for female avatars, 0.424 (95% CI [0.310, 0.538]) for male avatars, and 0.395 (95% CI [0.281, 0.510], OR = 1.03) for neutral avatars. These findings partially support the hypothesis: embodying a female avatar increased discomfort relative to male and neutral avatars, whereas male and neutral avatars did not differ.

4.1.3 Gendered Harassers Elicit Stronger Discomfort Than Neutral Harassers (H3)

We hypothesized that virtual harasser gender would influence perceived discomfort, with gendered virtual harassers eliciting higher predicted discomfort than neutral virtual harassers across participant and avatar conditions.

Table 4: **Interaction effects on predicted discomfort (H4–H7).** Effect sizes are reported as odds ratios (OR). P = participant gender, A = avatar gender, H = virtual harasser gender.

Hyp.	Interaction	OR	z	p
H4	Female P × Neutral H	2.15	2.74	0.006
	Female P × Male H	0.97	–0.11	0.913
H5	Female P × Male A	1.07	0.24	0.808
	Female P × Neutral A	0.70	–1.25	0.210
H6	Neutral A × Male H	0.45	–2.82	0.005
	Male A × Male H	0.64	–1.61	0.108
	Male A × Neutral H	0.62	–1.67	0.094
	Neutral A × Neutral H	0.80	–0.79	0.430
H7	Female P × Neutral A × Male H	2.19	1.97	0.049
	Female P × Male A × Male H	1.05	0.12	0.904
	Female P × Male A × Neutral H	1.66	1.27	0.204
	Female P × Neutral A × Neutral H	1.26	0.58	0.564

Virtual harasser gender showed a statistically significant main effect on predicted discomfort, shown in Fig. 5c. Averaged across participant gender and avatar gender, harassment enacted by female virtual harassers resulted in a higher predicted cumulative probability of high discomfort than harassment enacted by neutral virtual harassers (contrast: Female–Neutral = 0.396, SE = 0.082, $z = 4.82$, $p < 0.0001$, OR = 1.49). Similarly, male virtual harassers elicited higher predicted discomfort than neutral virtual harassers (contrast: Male–Neutral = 0.421, SE = 0.082, $z = 5.13$, $p < 0.0001$, OR = 1.52). No statistically significant difference was observed between female and male virtual harassers ($p > 0.05$). The estimated marginal predicted cumulative probabilities were 0.607 (95% CI [0.493, 0.721]) for female harassers, 0.632 (95% CI [0.518, 0.745]) for male harassers, and 0.211 (95% CI [0.096, 0.326]) for neutral harassers. These results support the hypothesis that gendered virtual harassers elicit higher discomfort than neutral harassers, with no reliable difference between male and female harassers, where $z = -0.31$, $p > 0.05$ and OR = 0.98.

4.2 Gender Dynamics in Virtual Harassment

To examine how combinations of gender-related factors shape perceived discomfort, we analyzed interaction effects among participant gender (P), avatar gender (A), and harasser gender (H). Tab. 4 summarizes the estimated interaction effects from the cumulative link mixed-effects model, including odds ratios (OR), test statistics, and significance levels. The following subsections describe the results for each interaction hypothesis.

4.2.1 Gendered Harassment Amplifies Discomfort for Female Participants (H4)

We examined whether the effect of virtual harasser gender on perceived discomfort differs between male and female participants. We expected male virtual harassers to elicit higher discomfort overall, particularly among female participants, while same-gender ha-

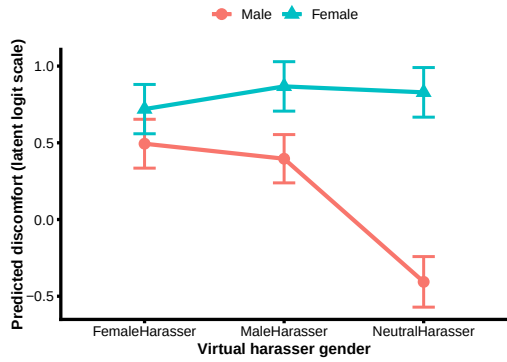


Figure 6: **Interaction between virtual harasser gender and participant gender.** Predicted discomfort estimates from the cumulative link mixed-effects model (latent logit scale), averaged across avatar gender. Error bars indicate 95% confidence intervals.

harassment may be experienced as less uncomfortable by male participants. Interaction effects were interpreted by examining how the predicted probability of reporting higher discomfort changed across conditions in the CLMM, averaged across avatar gender. To test whether these patterns differed by participant gender, we used difference-in-differences contrasts.

Results indicate a statistically significant interaction between virtual harasser gender and participant gender. In particular, the difference in predicted discomfort between male and neutral virtual harassers was significantly larger for female participants than for male participants (difference-in-differences estimate = 0.764, SE = 0.164, $z = 4.67$, $p < 0.0001$, OR = 2.15). A similar interaction pattern was observed for the contrast between female and neutral virtual harassers. As illustrated in Fig. 6, female participants exhibited consistently higher predicted discomfort in response to gendered virtual harassers, whereas male participants showed a marked reduction in predicted discomfort in the neutral harasser condition. In contrast, the difference between female and male virtual harassers did not vary reliably across participant genders (estimate = -0.031, SE = 0.280, $z = -0.11$, $p = 0.913$, OR = 0.97). These findings suggest that gendered harassment amplifies discomfort more strongly for female participants, while neutral harassment produces comparatively smaller gender differences.

4.2.2 Participant-Avatar Gender Matching Does Not Systematically Affect Discomfort (H5)

We examined whether perceived discomfort differed as a function of participant-avatar gender matching. We hypothesized that participants would experience different levels of discomfort depending on whether their avatar's gender matched their own. Specifically, stronger embodiment and identification may lead to higher discomfort in matched conditions, while mismatched avatars could either reduce discomfort through emotional distance or increase it through perceived vulnerability, depending on the context. To evaluate this hypothesis, we examined the interaction between participant gender and avatar gender within the cumulative link mixed-effects model. Interaction effects were interpreted by comparing predicted discomfort across avatar gender conditions as a function of participant gender.

The results did not reveal a statistically significant two-way interaction between participant and avatar gender. The interaction between female participants and male avatars was not significant (estimate = 0.068, SE = 0.280, $z = 0.24$, $p > 0.05$, OR = 1.07). The interaction between female participants and neutral avatars was also not significant (estimate = 0.068, SE = 0.280, $z = 0.24$, $p > 0.05$,

OR = 1.07). These results show that discomfort did not change in a consistent way when participants used avatars that matched or did not match their own gender. However, the model showed one significant three-way interaction. Female participants who used neutral avatars reported higher discomfort when they were harassed by male virtual harassers (estimate = 0.782, SE = 0.397, $z = 1.97$, $p < 0.05$, OR = 2.19).

These findings do not support the hypothesis as a general embodiment mismatch effect. Instead, any participant-avatar interaction appears to emerge only in specific harassment contexts rather than as a systematic pattern across conditions.

4.2.3 Avatar Identity Moderates the Impact of Harasser Gender (H6)

We examined whether the effect of virtual harasser gender depended on the gender of the embodied avatar. We tested this interaction using the cumulative link mixed-effects model.

The model showed a significant interaction between avatar gender and harasser gender. In particular, the effect of male virtual harassers differed across avatar conditions. When participants used neutral avatars, male virtual harassers elicited significantly lower discomfort compared to the reference condition (estimate = -0.792, SE = 0.281, $z = -2.82$, $p < 0.05$, OR = 0.45). The interaction between male avatars and male harassers was not significant (estimate = -0.444, $p > 0.05$, OR = 0.64). The interaction between male avatars and neutral harassers was marginal (estimate = -0.473, $p > 0.05$, OR = 0.62). Other avatar-harasser combinations were not statistically significant.

These results indicate that discomfort did not consistently increase in opposite-gender pairings. Instead, the impact of harasser gender varied depending on the avatar condition. The biggest difference appeared when male harassment was directed at neutral avatars.

4.2.4 Specific Gender Configurations Influence Discomfort (H7)

We further examined whether discomfort depends jointly on participant gender, avatar gender, and virtual harasser gender. This analysis tests whether the avatar-by-harasser interaction differs across participant genders.

The overall three-way interaction showed limited evidence of significance. Among the tested interaction terms, only the interaction involving female participants, neutral avatars, and male virtual harassers reached statistical significance (estimate = 0.782, SE = 0.397, $z = 1.97$, $p = 0.049$, OR = 2.19). All other three-way interaction terms were not statistically reliable ($p > 0.05$). These findings suggest that while certain specific gender configurations may modulate discomfort, the joint influence of all three gender dimensions does not produce a consistent or broad interaction pattern across conditions.

4.3 Interview Feedback

4.3.1 Participants' Perspectives on Harassment in Social Virtual Reality

Participants consistently expressed concerns about the prevalence and severity of harassment in social VR environments. Many felt that risks were more pronounced than in physical settings due to immersion and anonymity. A central concern was the lack of real-world accountability for harmful behavior. As P33 (female) explained: "There's no law enforcement in social VR. People can do whatever they want. Even if there's a penalty or punishment, it only exists in the virtual space. It doesn't carry over into real life". Some described direct encounters with harassment and the limited strategies available for response. P3 (male) recounted, "I experienced someone hitting me, and I just fought back to protect myself". Others highlighted verbal abuse and targeted hostility. As

P33(male) added, “*There are racist or harmful words, like someone following me and yelling insults. I usually just leave the platform directly*”. In response, some participants reported modifying their avatars to reduce unwanted attention. For example, P44 (female) noted, “*I usually choose a female avatar that reflects my personality, but sometimes I make it less detailed to avoid attention*”. Together, these accounts underscore participants’ perception that harassment in social VR is both persistent and insufficiently constrained by existing accountability mechanisms.

4.3.2 Avatar Gender Selection and Embodiment

Our quantitative data showed that 87% of participants preferred avatars that matched their self-identified gender. This choice was often motivated by comfort, familiarity, and a desire for accurate self-representation. As one female participant reflected, “*I have the same feeling when I experience harassment, whether I’m using a male, female, or neutral avatar. The behavior itself is what hurts*”. Another affirmed, “*I feel uncomfortable when they harass me, no matter which gender I am*”.

However, 13% reported choosing avatars of a different gender, most of whom were male participants citing aesthetic or social reasons. For example, P2 (male) explained, “*I would like to choose a female avatar and treat it as my daughter with a beautiful outlook, but I feel worse when harassment happens in my female avatar*”. Similarly, P13 (male) noted, “*Female avatars always have a better outlook and can easily join into a new social group*”. These reflections highlight how gender-swapping may offer social advantages yet simultaneously increase vulnerability to harassment, underscoring the trade-offs users face in avatar embodiment.

4.3.3 Additional Feedback and Design Suggestions

1. *Protecting Minors in Social VR.* Participants emphasized the importance of stronger protections for minors in shared spaces. Several expressed concern that children often interact in the same lobbies as adults, exposing them to inappropriate behavior. P19 (male) advised, “*I suggest isolating kids from social VR*”, while P34 (female) added, “*We need to protect kids from toxic and bad content. Probably kids can play only in educational versions like Minecraft*”. Others noted how minors respond under stress or conflict. For instance, P31 (female) observed, “*Kids are always in VR-Chat, they’re screaming sometimes*”. P12 (male) added, “*I know some kids who mute or tag each other when they feel angry, sometimes just for fun*”. P5 (female) concluded, “*This is a big problem because children should not be exposed to harassment or violent content*”. These concerns highlight the need for age-aware platform design, including restricted access, safer lobby segmentation, and child-focused content filters.

2. *Integrating LLM model for Real-Time Moderation.* Some participants proposed leveraging artificial intelligence to improve real-time moderation. P40 (male) suggested, “*If we could integrate AI into social VR, it might help reduce risks and create a safer environment. Features like customized personal boundaries and immediate reporting options would help*”. Such suggestions support the potential of AI-enhanced moderation to provide scalable, adaptive safety mechanisms that can detect and mitigate harmful behavior more effectively than current manual systems.

5 DISCUSSION

This study examined how participant gender, avatar gender, and virtual harasser gender jointly influence perceived discomfort in social virtual reality. Our results reveal that gendered harassment amplifies discomfort asymmetrically across participants. In particular, gendered virtual harassers elicited substantially higher discomfort among female participants compared to male participants, while neutral harassment produced comparatively smaller gender differences. These findings highlight the importance of considering both

user identity and avatar embodiment when evaluating harassment experiences in immersive environments.

5.1 Gendered Embodiment, Harasser Identity, and Social Context in Virtual Harassment

While a mismatch between participants’ self-identified gender and their avatar’s gender did not significantly affect their overall perception of harassment, our findings show that gender embodiment in virtual environments is a critical determinant of user experiences. Participants using female-presenting avatars reported higher levels of discomfort when exposed to harassment, particularly in sexualized contexts. This effect was consistent regardless of the participants’ real-world gender identity, suggesting that the visual and social signals associated with femininity may heighten vulnerability in virtual settings.

These results highlight that avatar embodiment influences more than visual representation. In immersive social VR environments, where users often experience intense feelings of presence and body ownership, avatars function as extensions of the self. Consequently, the gendered appearance of an avatar can affect how users interpret social encounters and respond emotionally to negative interactions. Nevertheless, not all participants attributed their discomfort to avatar gender. Some emphasized that the nature of the harassing behavior itself was the primary source of distress, regardless of how they were embodied. The gender presentation of the harasser also played a significant role in participants’ discomfort. Harassment initiated by male-presenting avatars was more likely to elicit strong negative reactions than similar behavior from female or neutral avatars. This aligns with prior research [32, 45] suggesting that users associate male avatars with greater threat potential, even in digital contexts.

The broader social environment also influenced how harassment was perceived. Virtual spaces with limited gender diversity or dominated by male-presenting avatars were often seen as less safe, potentially normalizing inappropriate behavior. In contrast, environments with more visible gender representation and diversity were associated with stronger feelings of safety and social accountability. These findings emphasize the importance of proactive design practices. While safety features such as muting and blocking remain essential, they primarily address incidents after they occur. A more comprehensive approach includes offering diverse avatar customization options, supporting gender-inclusive defaults, and creating interaction systems that promote respectful and equitable engagement across all users.

Future work should investigate how gender presentation, cultural context, and prior experiences interact to influence perceptions of safety and vulnerability in social VR. It is crucial to include the voices of non-binary, transgender, and gender-diverse users who are often underrepresented in immersive technology research but may face disproportionate risk. Their inclusion is essential to designing safer and more inclusive virtual environments.

5.2 Gendered Avatars and Varying Risks of Harassment

Building on these findings, our study reveals that female-presenting avatars are commonly targeted for harassment in social VR environments. Participants reported a higher frequency of unwanted attention, sexualized comments, and hostile interactions when embodying female avatars. This pattern reflects broader gendered dynamics in other online platforms, where femininity often draws both positive social engagement and harmful behavior [41, 67, 19]. In virtual reality and social VR, this effect is likely intensified by heightened presence and embodiment, where users experience their avatars as extensions of their bodies. This increased vulnerability is not limited to female-identifying users. Several male participants who chose female avatars for aesthetic or social reasons described heightened discomfort and a sense of vulnerability when targeted.

This suggests that harassment is shaped more by perceived avatar presentation than by the user's actual gender. Avatar appearance operates as a social signal that guides how others respond and ultimately affects users' sense of safety and empowerment. Harassment is not exclusive to female-presenting avatars. Participants also reported discomfort when using male or neutral avatars. Harmful behaviors such as verbal abuse, stalking, or mockery can affect any user, regardless of gender presentation. This highlights the complex interplay between avatar appearance, gendered expectations, and interpersonal dynamics in social VR.

These insights highlight the need for more advanced avatar design systems that consider both user expression and safety. Avatars are central to how users experience presence, construct social identity, and are perceived by others. Effective interventions should include flexible gender expression, controls over how avatars are perceived, and protections against behaviors that disproportionately target specific presentations. At the system level, behavioral monitoring, community moderation, and context-aware interventions can help reduce bias-driven interactions. Such measures can promote safer spaces for users across diverse identity groups.

5.3 Harassment in VR/AR/XR Affects Users Across the Gender Spectrum

Our results indicate that harassment in immersive environments disproportionately targets female-presenting avatars, while men also report serious emotional consequences, underscoring the need for inclusive safety measures. In virtual worlds, users can embody any identity, but harassers often focus on avatars perceived as vulnerable, regardless of the participant's actual gender. This decoupling between real-world identity and in-world appearance introduces complex social dynamics in harassment, where victimization is shaped as much by perceived vulnerability and visual cues as by the participant's offline gender. Male participants in our study reported discomfort when being harassed. While research shows that women face higher rates of harassment, particularly of a sexual nature [64], and men tend to underreport such incidents due to cultural pressures and gender norms, which can hinder support-seeking and exacerbate psychological harm [24, 35].

The embodiment effect in VR adds another layer of complexity. When participants inhabit avatars that strongly align with their self-identity, harassment can feel more personal and invasive. Conversely, mismatched avatars can either buffer emotional responses by creating a sense of detachment or heighten discomfort if the avatar is perceived as more vulnerable in the given social context. Prior research shows that avatar appearance strongly shapes users' perceptions of, and responses to, harassment [66, 65], suggesting that harassment prevention in VR should be sensitive to these identity-mediated dynamics. Beyond user experience, recent advances in active recognition have introduced motion-based classifiers such as HardenVR [65], proactive safety systems such as Puffer [33], real-time voice hate speech detection through systems such as SafeGuard [71], and model-driven approaches such as VS-Former [70] to detect and mitigate harassment in immersive environments automatically. From a design perspective, these findings challenge safety systems that are narrowly tailored toward one gendered experience of harassment. Features such as adaptive personal boundaries, customizable avatar visibility, and bystander intervention tools should be developed to address a broader range of harassment scenarios, including those affecting male and neutral avatars. Social norms interventions may help reduce the acceptability of harmful behavior across all gender pairings. Platform designers and policymakers should develop reporting systems that are accessible to all users, making it easier for individuals of all genders to report harassment. These systems should also include context-aware moderation tools that can detect both obvious and more subtle forms of abuse.

6 LIMITATIONS

Gender Diversity. This study included three gender categories: female, male, and neutral. This design choice reflects two considerations. First, many current VR platforms commonly provide these three options when users customize their avatars, making them the most widely available representations in existing systems. Second, the study aimed to group participants into clearly defined categories representing the largest user populations. However, this categorization does not capture the full diversity of gender identities. Many identities exist beyond these categories, and their perspectives are equally important. Collecting more detailed gender information may raise privacy concerns, as some participants may feel uncomfortable disclosing sensitive personal information, particularly in online or anonymous settings. In addition, recruiting participants from underrepresented gender groups can be challenging in smaller studies. Future research should explore more inclusive approaches to gender representation in VR studies and better capture the experiences of gender-diverse users.

Regional and Cultural Diversity. Although participants reported diverse racial backgrounds, all participants were based in the United States. Consequently, the findings may reflect cultural norms, values, and perceptions specific to U.S. users and may not generalize to other regions. Future work should include participants from a broader range of geographic and cultural contexts to better understand how social and cultural norms shape perceptions of harassment in social VR. Additionally, the harassment behaviors were presented in controlled experimental scenarios. While this design allows systematic comparison across conditions, real-world social VR interactions may involve more complex contextual cues.

7 AI USAGE DISCLOSURE STATEMENT

Social VR environment implementation, data collection, and analysis were conducted manually by the research team. Participants engaged with our custom VR study using Meta Oculus Quest 2 headsets. Post-study interviews were recorded using the Apple Recording app and Microsoft Teams. We organized and synthesized qualitative results using Miro [40]. To streamline writing, we used large language models (LLMs), specifically GPT-5, to refine wording, improve clarity, and enhance readability. All substantive analytical decisions, thematic codes, and interpretive claims remain the responsibility of the human authors. Interpretation of findings and the crafting of our discussion were primarily human-led. We disclose this AI usage as part of our commitment to transparency in AI-mediated scholarship.

8 CONCLUSION

This study demonstrated that harassment in social VR is significantly influenced by three factors: the type of harassment, the embodiment of avatar users, and the gender of the harasser. Sexual harassment elicited the highest levels of discomfort, embodiment in female avatars increased feelings of vulnerability, and male-presenting harassers provoked the strongest negative reactions. Through a mixed-methods study in a custom-built VR environment, we integrated quantitative ratings with qualitative insights to capture both the intensity and context of these experiences. By employing standardized NPC behaviors and randomized layouts, we established a controlled and ethically responsible approach for investigating sensitive interactions. Beyond methodological contributions, the findings underscore the need for stronger prevention mechanisms, more inclusive avatar design, and robust safeguards to ensure safer and more equitable virtual environments. Future research should extend this work by including more diverse populations and exploring additional identity factors to deepen our understanding of harassment in immersive environments.

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